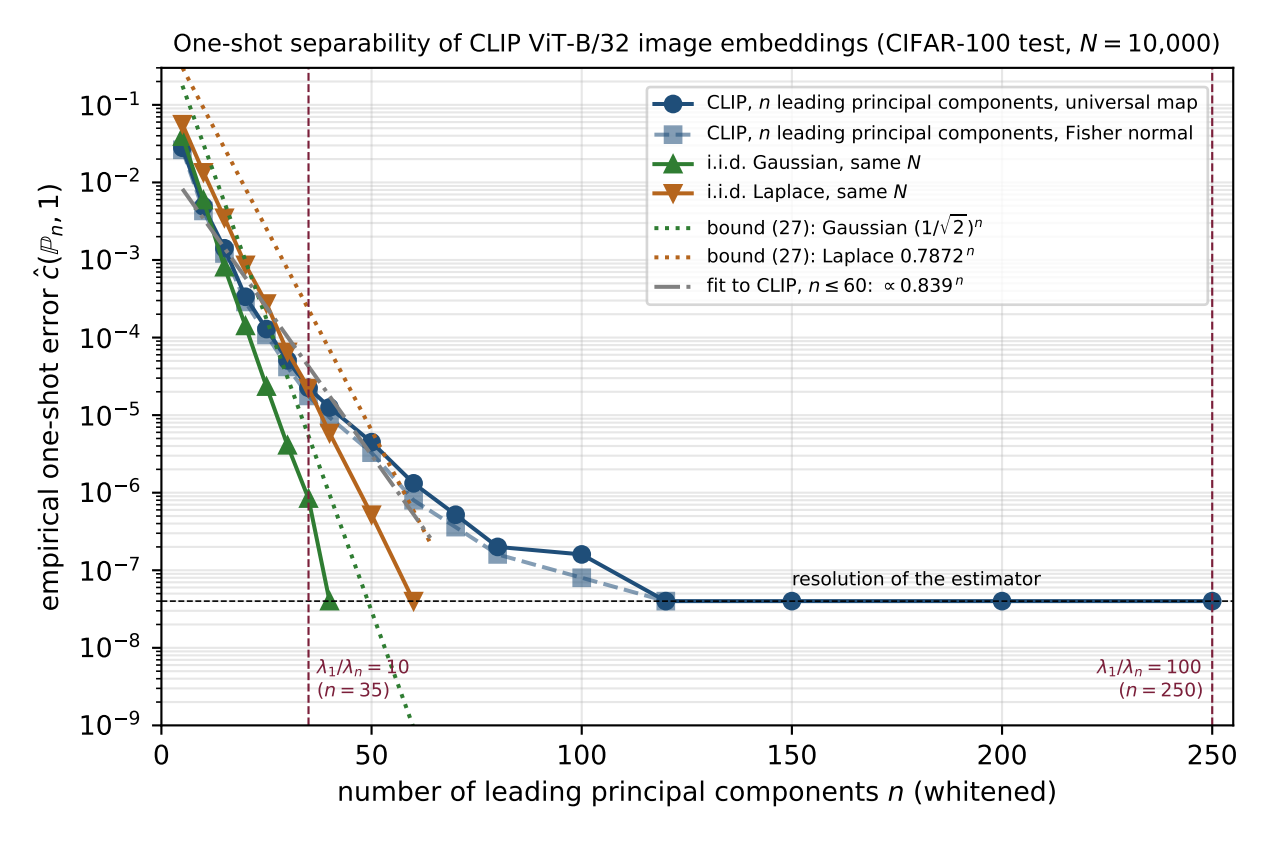
One-shot separability of CLIP ViT-B/32 image embeddings (CIFAR-100 test, $N = 10{,}000$)
empirical one-shot error $\hat{c}(\mathbb{P}_n, 1)$
number of leading principal components n (whitened)
CLIP, n leading principal components, universal map
CLIP, n leading principal components, Fisher normal
i.i.d. Gaussian, same N
i.i.d. Laplace, same N
bound (27): Gaussian $(1/\sqrt{2})^n$
bound (27): Laplace 0.7872^n
fit to CLIP, $n \le 60$: $\propto 0.839^n$
resolution of the estimator
$\lambda_1/\lambda_n = 10$
$(n = 35)$
$\lambda_1/\lambda_n = 100$
$(n = 250)$